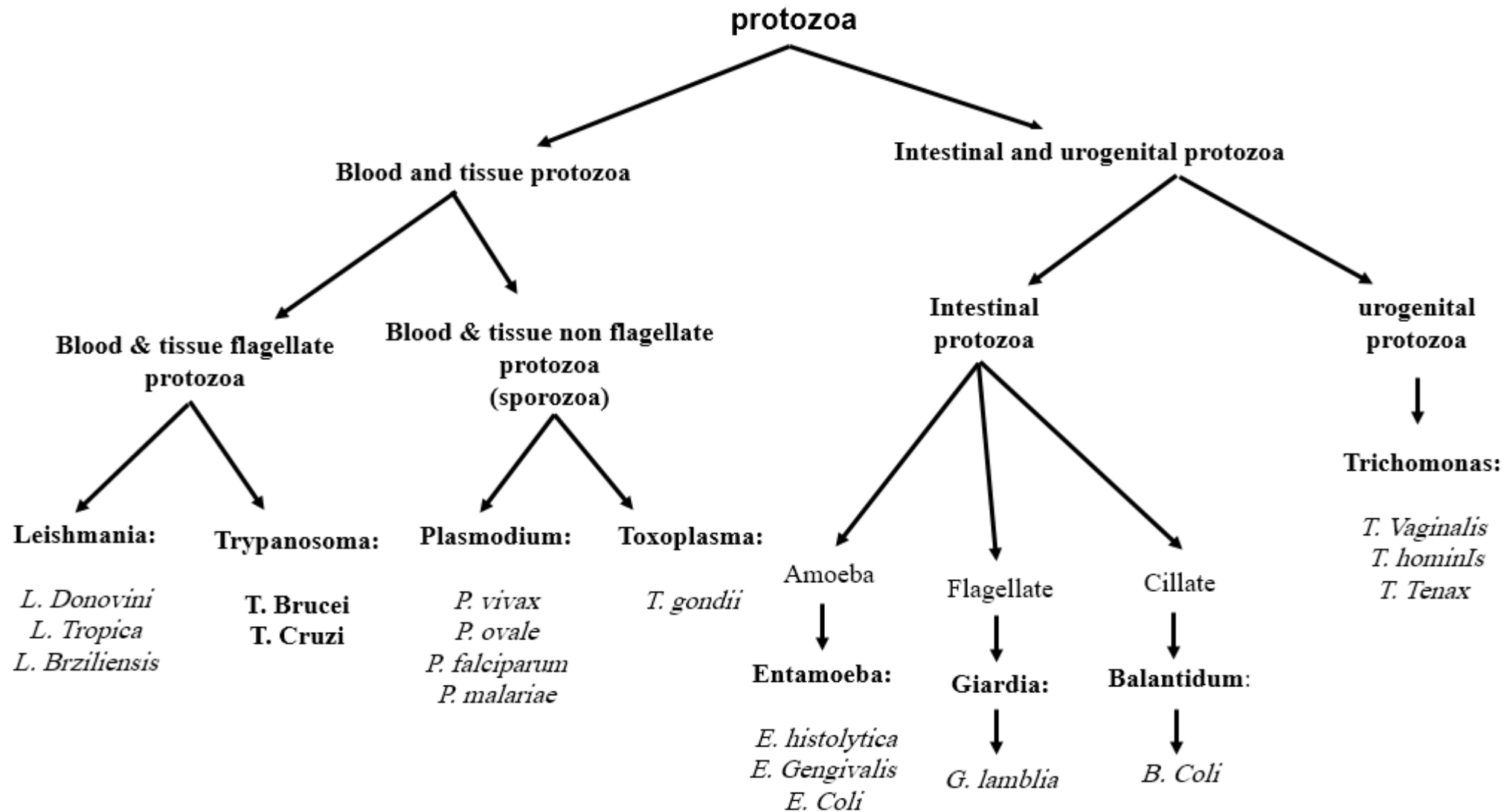


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First course Parasitology - Protozoa

Lecture 2: General Protozoa



The most important classification of some medical protozoa according the site of infection

Protozoa	Habitat	Infective & route of entry	Disease	Reservoir host	diagnostic stage
<i>Entamoeba histolytica</i>	Large intestine	Cystic stage orally	Amoebic dysentery and extra intestinal	Humans	Cyst and trophozoite in patients stool
<i>Balantidium coli</i>	Large intestine	Cystic stage orally	Balantidium dysentery	Animals (pigs)	Cyst and trophozoite in patients stool
<i>Giardia lamblia</i>	Small intestine	Cystic stage orally	Diarrhea, malabsorption and fatty diarrhea (steatorrhea)	Humans	Cyst and trophozoite in patients stool
<i>Trichomonas vaginalis</i>	Vaginal and urethra	Trophozoite by STD	Vaginitis or urethraitis	Humans	Trophozoite stage from Vaginal or urethra
<i>Plasmodium spp.</i>	Liver and RBC	Sporozoite skin	Malaria	Humans	Ring stage from peripheral blood
<i>Toxoplasma Gondii</i>	obligate Intracellular	Oocyst tissue cyst orally	Congenital toxoplasmosis	Animals	Tachzoite and tissues cyst from bone marrow

<i>Leishmania donovani</i>	Reticuloendothelial system	Promastigote skin	Kala-azar	Animals & Humans	Amastigote
<i>Leishmania tropica</i>	Skin	Promastigote skin	Cutaneous leishmaniasis	Humans & Animals	Amastigote
<i>Leishmania braziliensis (espundia)</i>	Naso or oral mucosa	Promastigote skin	Mucocutaneous, leishmaniasis	Humans & Animals	Amastigote
<i>Trypanosoma brucei</i>	blood., lymph node and CNS	Metacyclic-trypomastigote skin	Sleeping sickness	Animals & Humans	Trypomastigote
<i>Trypanosoma cruzi</i>	Macrophage, bone marrow never heart and colon	Metacyclic-trypomastigote skin	Chagas and congenital disease	Animals & Humans	Trypomastigote

The most important protozoa species with their infective stage, habitat, disease, reservoir host, and diagnostic stage

Types of hosts

Host: all species of parasites need either one host or two hosts to complete their life cycle in order to reach the infective stage. Sometimes two or more intermediate hosts are needed.

There are different types of hosts:

Definitive (final) host: some species of parasites need one host to complete their life cycle like protozoa e.g. *Amoeba* (***Entamoeba Histolytica***), ***Giardia lamblia*** and helminths e.g. ***Enterobius Vermicularis***, ***Hymenolepis Nana***. This is the human body or animal in which the adult parasite lives and undergoes sexual reproduction and is called definitive host, and include protozoa like ***plasmodium*** in female anopheles, *Toxoplasma gondii* in the cat and leishmania in human body .

Intermediate hosts:

some parasites need two hosts to complete the life cycle, final and intermediate hosts, the parasite live in this host as an immature phase (larva stage) and grow by asexual reproduction like protozoa e.g. ***Plasmodium species*** in human, helminths e.g. ***Schistosoma species*** in snail.

Reservoir host:

is an animal form of infection may be passed to other species and is not affected by the infection (they are usually wild animals), such animals help to maintain the cycle of parasites in nature and act as potential source for human infection with this parasite, like protozoa e.g. ***Leishmania species*** in animals. Helminths e.g. ***Trichinella spiralis*** in pork. These types of hosts are called carrier hosts in the human like protozoa e.g. ***Entamoeba histolytica***.



All species of parasites need one or two host to complete their life cycle to reach the infective stage.

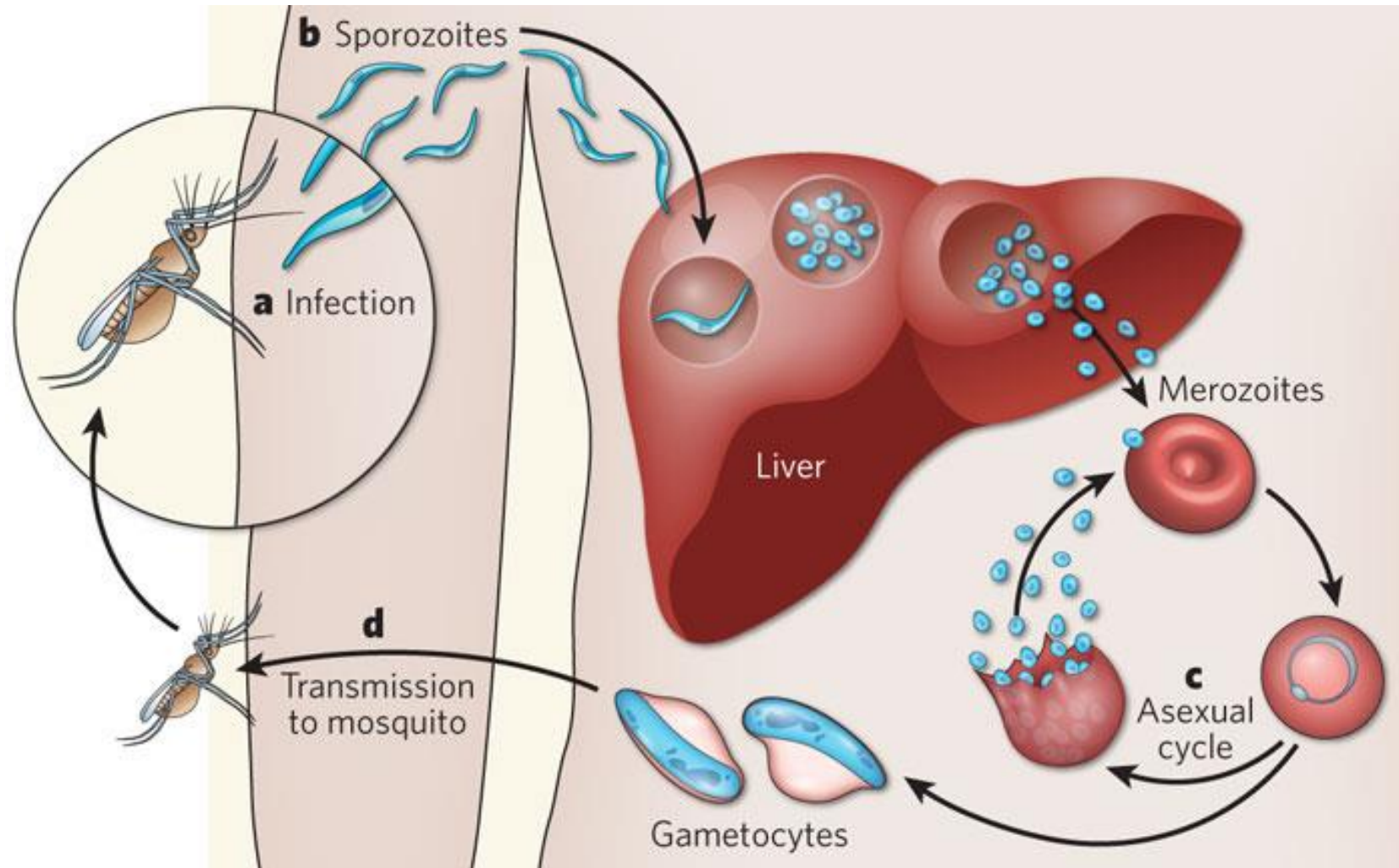
There are two kinds of hosts:

One host (direct life cycle): Like intestinal protozoa e.g. ***Entamoeba histolytica*** , ***Giardia lamblia***, and ***Balantidium coli***, and for helminths like ***H. Nana***. All of them require only one host as the human body, is characterized by different stages, their life cycle in human host is trophozoite, and cyst stage as a diagnostic stage and the infective stage



Two hosts (indirect life cycle) like:

I. Protozoa: like *Plasmodium spp.*, first host is the human body as an intermediate host and the reproduction is by asexual reproduction, second host is the female anopheles as the definitive host and the reproduction is by sexual reproduction, the infective stage is sporozoite, and the diagnostic stage is ring stage.

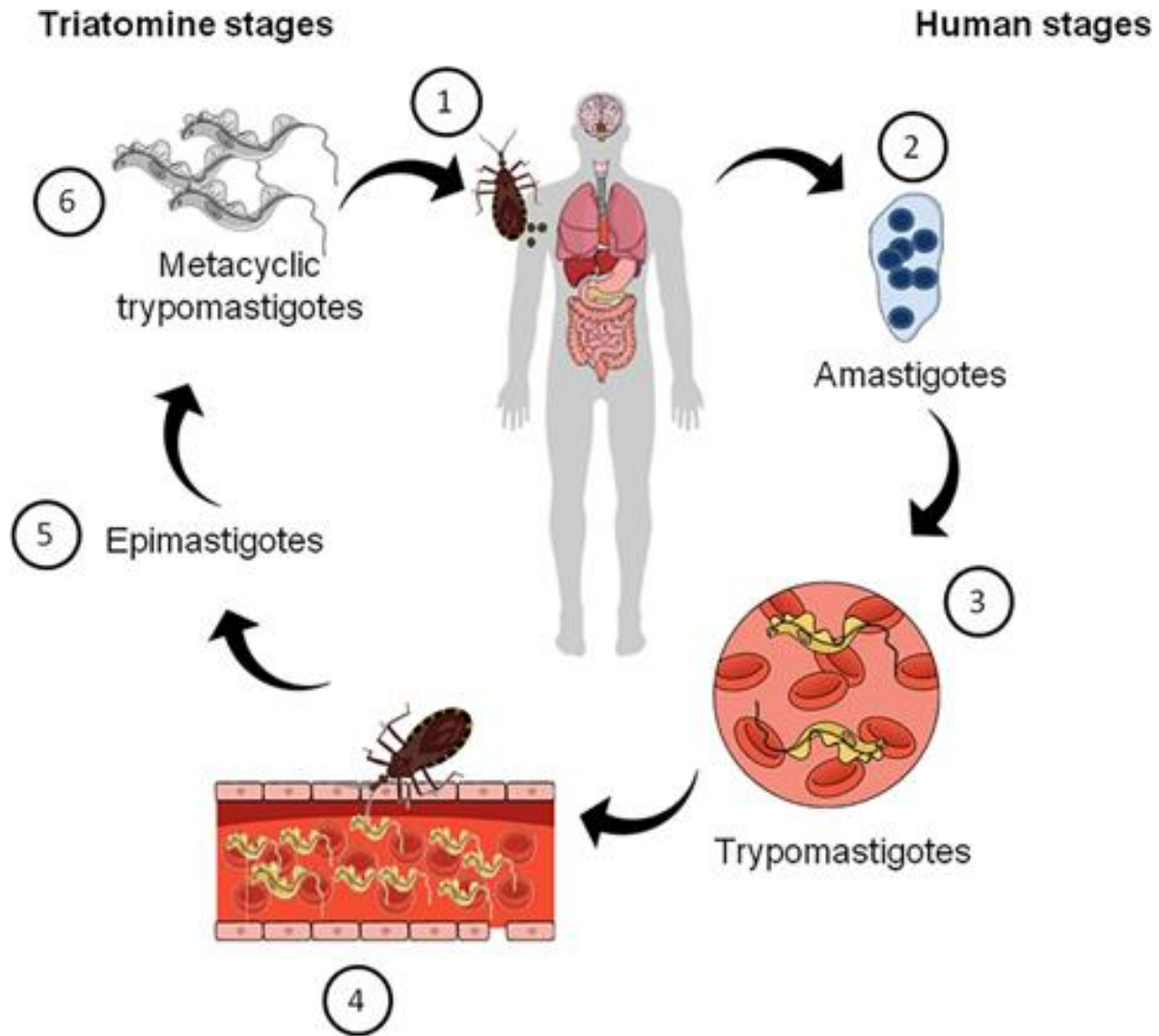


Life cycle of *plasmodium* species



Trypanosoma Brucei:

First host is the human body as definitive host and reproduction by a sexual reproduction, second host is the tse-tse fly as an intermediate host, and the reproduction by a sexual reproduction. Infective stage are metacyclic trypomastigote and diagnostic stage are trypomastigote

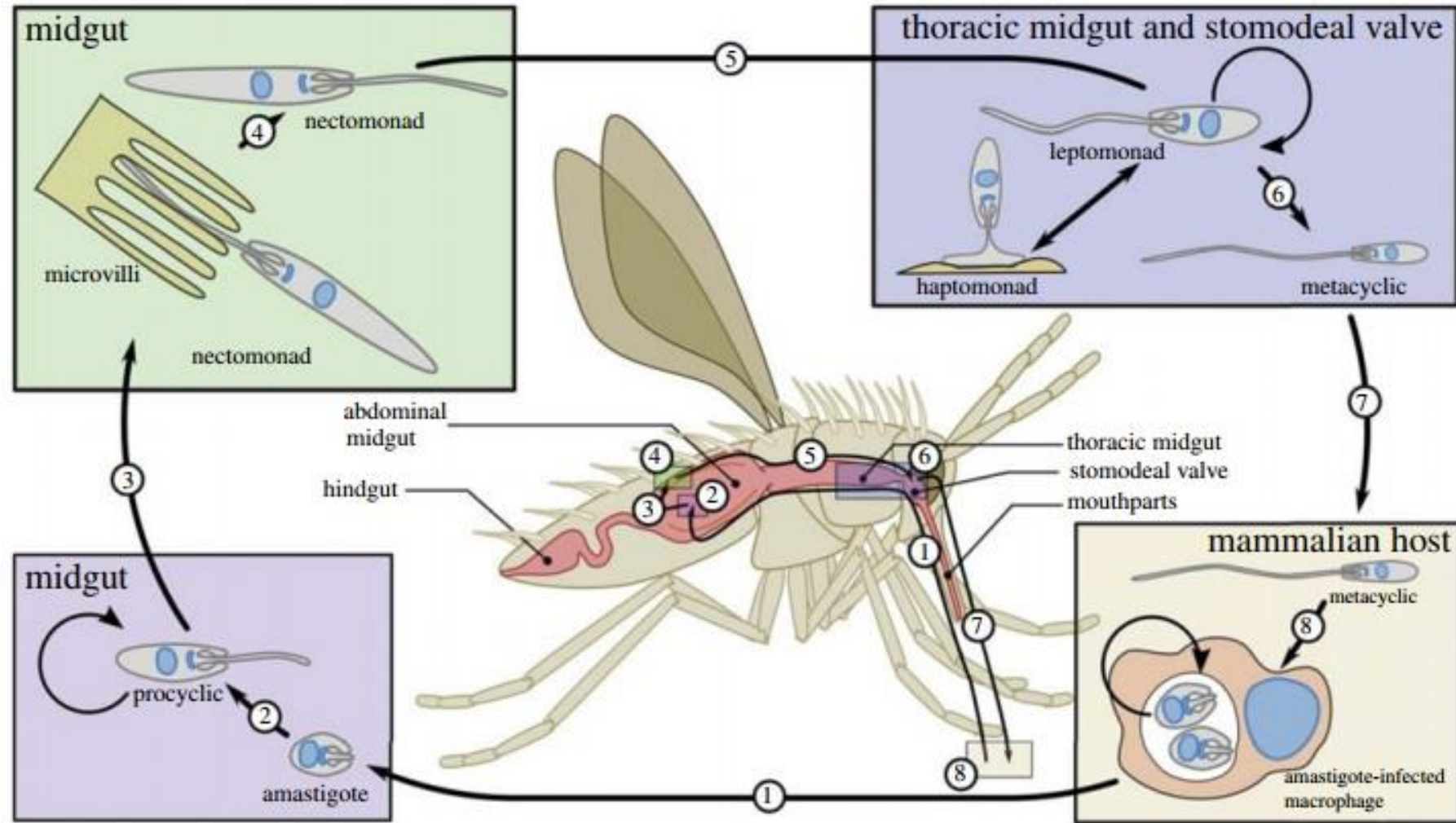


Life cycle of *Trypanosoma Brucei*



Leishmania spp.

First host is the human body as the definitive host and the reproduction by a sexual reproduction, and second host is the female sand fly as an intermediate host and the reproduction by a sexual reproduction. Infective and diagnostic stages are promastigote.

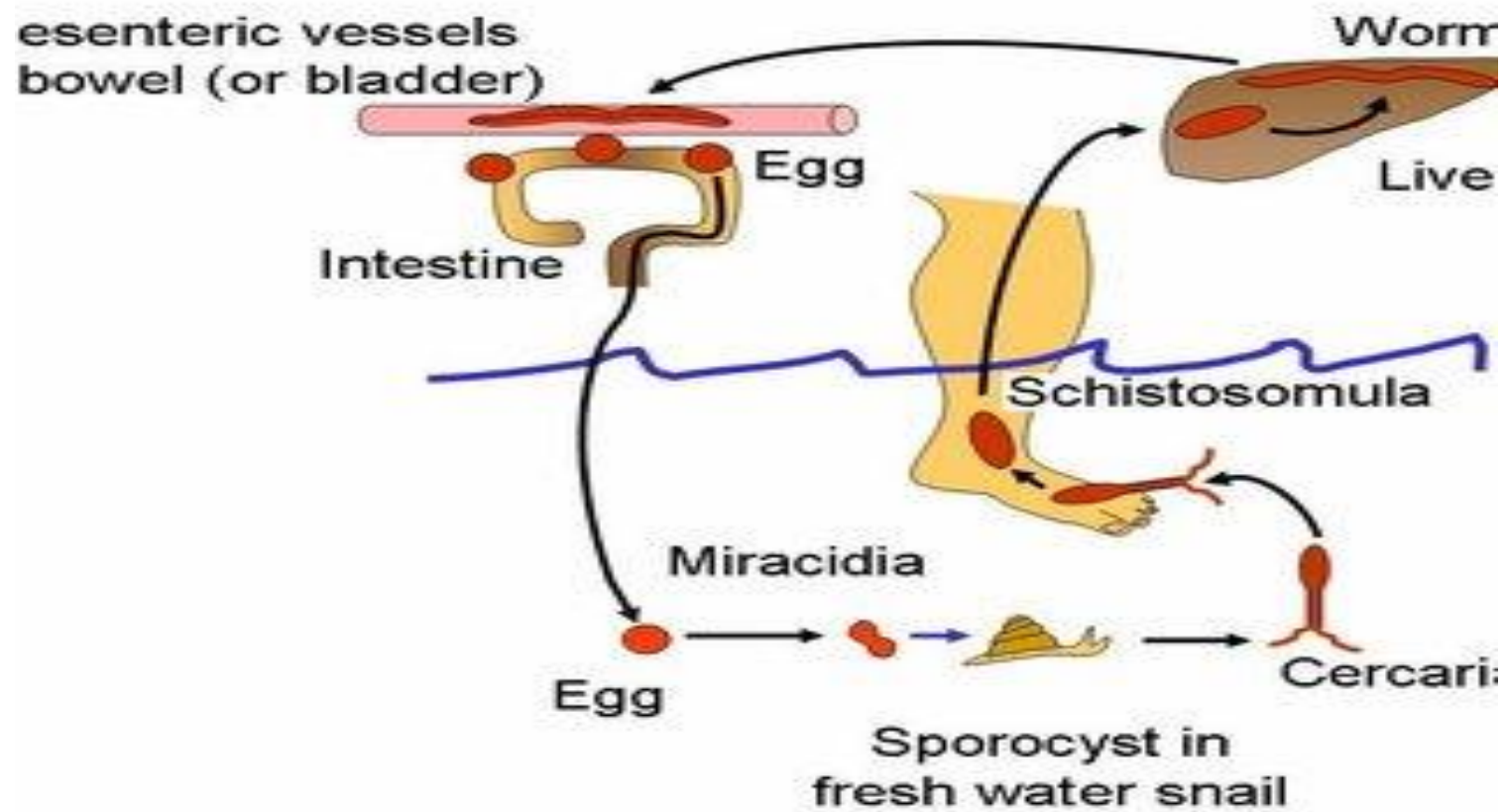


Life cycle of *Leishmania* spp.



2. Helminths:

e.g. ***schistosoma*** , first host is human body as definitive host and reproduction by sexual reproduction, and the second host is snail as intermediate host and reproduction by a sexual reproduction, the infective stage is cercaria in water and diagnostic stage is eggs in patients stool or urine according to the species.



Life cycle of *Schistosoma* species



Some times the helminths need two or more intermediate host e.g. ***Paragonimus westermani*** and ***Diphyllobothrium latum***, Human body as definitive host, the snail as first intermediate host, and fish species as second intermediate host.

Infective stage is metacercaria with young flukes for ***Paragonimus westermani*** and plerocercoid larva in infected fish for ***D. latum***



Pathogenicity

The pathogenesis of parasitic infection (helminths or protozoa) may take many forms: acute, subacute, chronic and latent or recurrent diseases depends upon several factors, such as the density of parasites, and site of infection, as well as specific mechanisms of tissue damage.

The density of infection depends on the species of the parasite, the age and condition of the host, inter and intra specific competition of the parasite, in addition to other factors.



Pathogenic mechanisms by which parasites can damage the host include the following:

1. **Destruction of red blood cells** e.g. *Plasmodium* (malaria).
2. **Destruction of skin tissues** e.g. **skin ancylostomiasis** (*hookworm*) by penetration of filariform larva or **skin schistosomiasis** by penetration of cercaria in water.
3. **Lytic necrosis :-** Destruction of organs by parasitic enzymes produced by some species of protozoa and helminths e.g. protozoa liver as in **hepatic abscess amoebiasis** (*Entamoeba histolytica*) in human liver and lyses intestinal mucosa, and lyses the tissue layer of wall of large intestine and cause deep ulceration as in **intestinal amoebiasis** (*Entamoeba histolytica*), helminths like cercaria of *schistosoma*, Filariform of *hookworm* when penetrate the human skin.

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4. **Trauma:** when adult of *hookworms* (*Ancylostoma duodenale*, and *necator americanus*) attachment on jejunal mucosa leads to traumatic damage of villi and bleeding at the site of attachment
 5. **Nutritional robbing:** as in helminth causing anemia in case of **intestinal ancylostomiasis** or **necatoriasis**.
 6. **Inflammation:** due to secondary bacterial infection.
 7. **Inflammation and obstruction of helminth:** the lymphatic vessels causing edema as in case of **elephantiasis** (filaria worm - *Wuchereria bancrofti*). And sandy patches or **urinary bladder granuloma** in case of *Schistosoma haematobium*.

Filariasis

Injects microfilaria





Allergic manifestations: clinical illness may be caused by host immune response to parasitic infection e.g. anaphylactic shock in rupture of hydatid cyst mainly in **brain hydatidosis** and **eosinophilic pneumonia** in ***ascaris* infection**

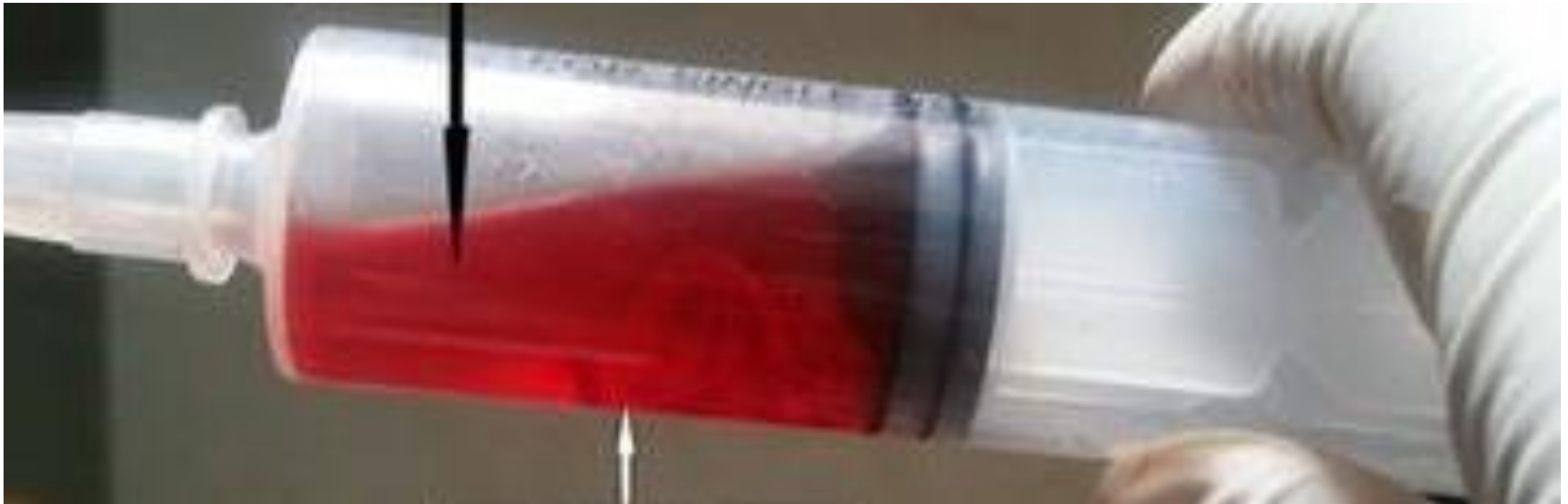
Neoplasia: A few parasitic infection have been shown to lead to malignancy e.g. bile duct carcinoma (due to liver fluke), urinary bladder cancer due to infection by ***S. haematobium***.

Inflammation: (protozoa) of the brain fluid and spinal cord, mediated by released toxins as in the case of **sleeping sickness** (*Trypanosoma brucei*).



Intermediate host, tse-tse fly (sleeping sickness)

Hematuria: in helminths when eggs of ***Schistosoma haematobium*** damage the wall of the urinary bladder.



Macroscopic examination of urine due to *S. haematobium*

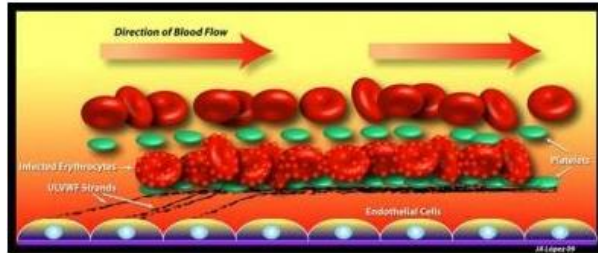
Physical obstruction: It is an obstruction of intestine due to accumulation of adult *Ascaris lumbricoides* in cavity of intestine

Adult *ascaris lumbricoides*
in human small intestine

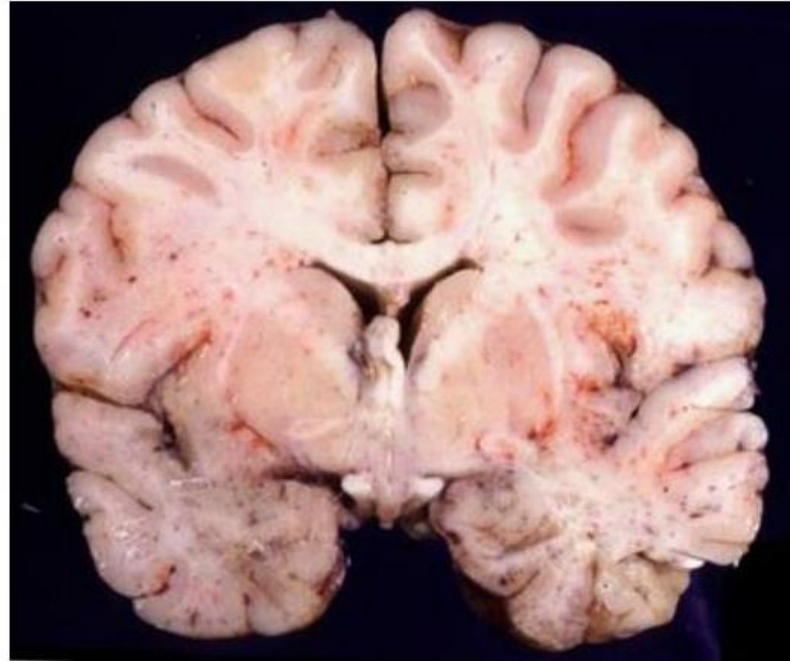


A

And due to ***Plasmodium falciparum*** (malaria) causing blockage of Brain capillaries in cerebral malaria.



B



Section of a brain of fatal case
of cerebral malaria



Treatment

Treatment means complete killing, control, or destruction of parasites or its stages inside the human body.

Specific chemotherapeutic agents (antiparasitics) are used against each parasite, although some have side effects, tapeworm infections are usually treated with an oral medication such as **praziquantel**, which paralyses the adult tapeworm and then passes out the body with stool.

The fact, which necessitates that their use must be used under medical supervision. Treatment by **nifurtimox** can be done by stimulation of the infective cell to produce special substances to kill the parasites like H₂O₂ as in the case of **Chagas disease (*Trypanosoma cruzi*)**.



Common treatments for roundworm diseases include **mebendazole** and **albendazole**.

The symptoms typically begin to improve after a few weeks of treatment. Antiprotozoal **metronidazole (flagyl)** is the most common treatment for intestinal protozoa like **giardiasis**, **balantidiasis**, and genital protozoa like ***trichomonas vaginalis***, it works by damaging protozoal DNA.

Some parasitic diseases can be treated by **surgical removal of cysts** and tumors caused by parasites that cannot be cured by **chemotherapeutic** (antiparasitics) agents' such as **hydatidosis**.



Antiprotozoal drug agent that kills or inhibits the growth of the protozoal cells are known as protozoans.

Protozoans treat all protozoal disease which include intestinal protozoa, blood and tissue protozoa, and urogenital protozoa. Commonly used as **oral antiprotozoal drug or sometime intramuscular (IM).**

The action of antiprotozoal drug agent on the infection is complex and are not fully understood .

Some of them may interfere with reproduction of protozoa or damage protozoal DNA to limit the spread of the infection

Ex. **Antimalarial drug** like **chloroquine** kill or inhibit the growth of *plasmodium* by affecting different stage of *plasmodium* in human life cycle, they are used both to treat and prevent malaria .



Prophylaxis

- Prevention of contamination of food and water by patient's stool (infective stage),
- Detection and treatment of carriers and their exclusion from feed handling occupations will help in limiting the spread of infective stage of disease.
- Health education and inclusion of healthy personal habit helps in control.
- Proper disposal of waste and feces.
- The practice of personal hygiene like hand washing before cutting and proper disposal of diapers.
- Prevention is the same as for sexually transmitted diseases (STD) like ***vaginal trichomoniasis***.



Control

Is based on early diagnosis and treatment of cases to reduce the reservoir or carrier host infection. Control or killing the insect by biological or chemical methods, Integrated insecticidal spraying to kill or reduce the infected insect.

The distraction of animal reservoir host which causes the zoonotic disease. Personal prophylaxis by using anti-infected insect, like using thick clothing, bed net, window mesh etc.

Prevention by vaccination



Prevention of parasitic disease can be done by forming preventive obstacles against spreading of parasites and their entrance to the human body.

These are carried out by practical application of biological control and prophylaxis in order to destroy the transmitter hosts.

Best ways of prevention can be by sanitary disposal of human stool using the proper health procedures or by treatment with chemical agents and not use it as a fertilizer directly.

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- Destroying intermediate hosts, reservoir hosts and insects which transmit parasites by using chemical or biological control.
 - Health control of drinking water by filtration and disinfection with chemicals to kill parasitic stages in water.
 - Vaccination against parasitic diseases.
 - Cleaning and disinfection of fresh vegetables before eating.
 - Treatment and isolation of infected persons.
 - The use of health education programs about prophylaxis from parasitic disease.



The following steps are followed to control parasitic diseases:

- Reduction of source of infection by treatment.
- Health education in order to prevent spreading of infection.
- Building of biological obstructs against transmission of parasitic diseases.



Vaccination

At general no effective vaccine for human has so far been developed against parasites, due to their complex life cycles, adaptive responses, and antigenic variation.

Great progress has been made in identifying protective antigens in malaria and some other infections with a view to eventual development of prophylactic vaccines



Thank you for listening, any questions?